

Shubham Kaushik

Data Systems Researcher | Software Engineer

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Build storage engine internals in C++ and Rust: merge policies, memory layouts, and learning-based database tuning.

Four years of industry engineering and three years of research; published at VLDB, ICDE, TPCTC and DBTest.

EXPERIENCE

Ph.D. Researcher, Smart & Scalable Data Systems Lab @ Brandeis University, MA, USA Jan 2024 - Present

- Design and evaluate LSM-based storage engine internals such as merge policy, layouts, and write-buffer design.
- Apply learning-based methods to storage-engine configuration, from offline tuning to runtime adaptation.
- Develop benchmarking suites used to evaluate key-value stores under dynamically shifting workloads.

Software Engineer, Scalability & Infrastructure Team @ Kwalee, India Mar 2022 - Aug 2022

- Designed and shipped server-side leaderboard and loot-box systems integrated into three game releases.

Engineer - Information Security, Information Security Automation @ FIS Global, India Jun 2021 - Mar 2022

- Built an asynchronous IP-scanner to scan over 100K+ internet endpoints, replacing the team's manual spot-checks.
- Authored and merged integrations for DB2, GenericSQL and FireMon into Palo Alto's Cortex XSOAR [[demisto/content](#)].

Project Engineer, Cloud Computing and Big Data Teams @ Wipro Limited, India Jul 2018 - Jun 2021

- Built a microservice that pseudonymises PII in CSV, Excel and mainframe files, replacing a manual review of 1K+ files.
- Designed a real-time agent that flags anomalous hosts across 500+ enterprise systems using k-means clustering.

PROJECTS [SELECTED]

FluidLSM — self-tuning storage engine [[blog](#)] 2025 - Present

- Designing an LSM engine (prototype on RocksDB) that retunes itself as workloads shift, using reinforcement learning to search its configuration space, with an LLM acting as a DBA over prior tuning runs to narrow the search.

RangeReduce — query-driven compaction [[PDF](#)] [[GitHub](#)] ICDE 2026

- Designed a range query driven compaction strategy that reduce range query latency and the need for future compactions.
- Improves latency up to 18%, compaction debt up to 90%, data movement up to 12%, and space amplification up to 20%.

Tectonic → **TexBench** — key-value benchmarking suite [[GitHub](#)] VLDB 2026, TPCTC 2025/26

- Built *Tectonic*, a Rust key-value workload generator replacing YCSB: 2× higher throughput, 84% lower memory footprint.
- Developed *TexBench*, an LLM-powered layer on *Tectonic* that lets users specify benchmark workloads in plain English.
- Designed a Domain-Specific Language that restrict LLMs to generate eight edit commands to produce valid benchmarks.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust

Databases: RocksDB, PostgreSQL, Redis

Systems & Performance: Linux, perf, gdb, CMake, Docker

Machine Learning: Reinforcement Learning (RL), LLMs, DSL, Retrieval-Augmented Generation (RAG), PyTorch

PUBLICATIONS [SELECTED]

TexBench: Harnessing LLMs for Efficient Key-Value Benchmarking — **Kaushik** et al. TPCTC 2026

TexBench: A Unified Benchmarking Suite for Shifting Workloads (demo) — Chanda, **Kaushik** et al. VLDB 2026

RangeReduce: Query-Driven Compaction in LSM-Trees — **Kaushik** et al. ICDE 2026

Tectonic: Bridging Synthetic and Real-World Workloads — Alexander H. Ott, **Kaushik** et al. TPCTC 2025

Anatomy of the LSM Memory Buffer: Insights & Implications — **Kaushik** and Sarkar DBTest 2024

EDUCATION

Ph.D. in Computer Science (GPA: 3.98/4.0) | **Brandeis University** Jan 2024 - Present

M.S. in Computer Science, w/ Data-Centric Computing (GPA: 3.88/4.0) | **Boston University** Sep 2022 - Dec 2023

B.Tech. in Computer Science & Engineering | **Maharshi Dayanand University** Jul 2014 - Jun 2018

TALKS [SELECTED & INVITED]

RangeReduce: Query-Driven LSM Compactions @ **MiDAS Seminar**, Boston University, USA 04/2026

Range Queries and LSM-Trees: From Foes to Friends @ **Salesforce**, USA 10/2025

SERVICE

Program Committee, **VLDB 2027** · VLDB 2026 Shadow PC · Web Chair, NEDB Day 2025 · Reviewer, IEEE Big Data

OUTREACH

Volunteered at **VLDB 2026** · Judged and mentored at **HackMIT** · Mentored at **BostonHacks**